

Unitriangularity of decomposition matrices

Simple adjoint finite reductive groups of exceptional type

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Introduction and motivation

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- $\mathbf{K} = \text{Frac}(\mathcal{O})$ field of characteristic zero (“big enough for G ”),
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Definition

The ℓ -**decomposition matrix** $D = (D_{V,W})_{V \in \text{Irr}_{\mathbf{K}}(G), W \in \text{Irr}_{\overline{\mathbb{F}_\ell}}(G)}$ is given by

$$D_{V,W} := \dim \left(\text{Hom}_{\overline{\mathbb{F}_\ell}[G]}(V^\mathcal{O} \otimes_{\mathcal{O}} \overline{\mathbb{F}_\ell}, W) \right).$$

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Thus, the 3-decomposition matrix of S_3 is $D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$.

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- $P^{\mathcal{O}} \otimes_{\mathcal{O}} K$ a $K[G]$ -module,
- $P^{\mathcal{O}}$ an $\mathcal{O}[G]$ -lattice s.t. $P^{\mathcal{O}} \otimes_{\mathcal{O}} \overline{\mathbb{F}}_{\ell} \cong P$,
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There is a bijection between $\text{Irr}_{\overline{\mathbb{F}}_{\ell}}(G)$ and the set of projective indecomposable $\overline{\mathbb{F}}_{\ell}[G]$ -modules up to isomorphism, sending W to P_W .

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$$P_W^{\mathcal{O}} \otimes_{\mathcal{O}} \mathbf{K} = \bigoplus_{V \in \text{Irr}_{\mathbf{K}}(G)} D_{V,W} V = \bigoplus_{V \in \text{Irr}_{\mathbf{K}}(G)} \langle V, P_W^{\mathcal{O}} \otimes_{\mathcal{O}} \mathbf{K} \rangle V.$$

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Idea behind decomposition matrices

Ordinary modules + decomposition matrices $\approx$ modular modules.

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Example

If $G = S_3$, the 3-decomposition matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ is unitriangular.

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Conjecture (Geck 1990)

The decomposition matrix of a finite reductive group in non-defining characteristic is lower-unitriangular (w.r.t. a suitable ordering).

Finite reductive groups and their representation theory

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The **finite groups of Lie type** or the **finite reductive groups** are the groups of the form $\mathbf{G}^F := \{g \in \mathbf{G} \mid F(g) = g\}$.

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Example

Take $\mathbf{G} = \mathrm{GL}_n(\overline{\mathbb{F}}_p)$ for $n \in \mathbb{Z}_{>0}$ and

$$\begin{aligned} F : \mathbf{G} &\rightarrow \mathbf{G} \\ (a_{ij})_{1 \leq i, j \leq n} &\mapsto (a_{ij}^q)_{1 \leq i, j \leq n}, \end{aligned}$$

then $\mathbf{G}^F = \mathrm{GL}_n(\mathbb{F}_q)$.

Ordinary representations

Fix an F -stable Borel subgroup $\mathbf{B} \leq \mathbf{G}$ and an F -stable maximal torus $\mathbf{T} \leq \mathbf{B}$. We write $(\mathbf{G}^*, F^*)$ for the pair in duality with $(\mathbf{G}, F)$.

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Theorem (Deligne–Lusztig 1976)

There is a partition

$$\mathrm{Irr}_K(\mathbf{G}^F) = \bigsqcup_s \mathcal{C}(\mathbf{G}, s),$$

where s runs over a set of representatives of F^ -stable $\mathbf{G}^*$ -conjugacy classes of semisimple elements in $\mathbf{G}^*$.*

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The series $\mathcal{E}(\mathbf{G}, 1)$ is called the **unipotent** series.

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Theorem (Broué–Michel 1989)

Fix a prime number $\ell \neq p$. Assume that F is a Frobenius map. Let $s \in (\mathbf{G}^)^{F^*}$ be a semisimple ℓ' -element. Define*

$$\mathcal{E}_{\ell}(\mathbf{G}, s) := \bigsqcup_t \mathcal{E}(\mathbf{G}, st),$$

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The ℓ -decomposition matrix of the unipotent ℓ -blocks of $\mathbf{G}^F$ is lower-unitriangular if p is good for $\mathbf{G}$ and ℓ is very good for $\mathbf{G}$ (good + condition related to $Z(\mathbf{G})$).

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Strategy and results

Our setting

We fix:

- $\mathbf{G}$ a connected reductive **simple** group of **adjoint** type defined over $\overline{\mathbb{F}}_p$ for p a **good** prime for $\mathbf{G}$,
- $F : \mathbf{G} \rightarrow \mathbf{G}$ a **Frobenius** map,
- $\ell \neq p$ a prime number,
- $(\mathbf{G}^*, F^*)$ in duality with $(\mathbf{G}, F)$,
- $(\mathbf{K}, \mathcal{O}, \overline{\mathbb{F}}_\ell)$ a splitting ℓ -modular system for $\mathbf{G}^F$ and
- $\mathcal{B} := \mathcal{E}_\ell(\mathbf{G}, s)$ for a semisimple F^* -stable ℓ' -element $s \in \mathbf{G}^*$.

Goal

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Step 1. Find the number n of irreducible $\overline{\mathbb{F}}_\ell[\mathbf{G}^F]$ -modules in $\mathcal{B}$.

Step 2. Find candidates $V_1, \dots, V_n \in \text{Irr}_K(\mathbf{G}^F)$ in $\mathcal{B}$.

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Steps 1 and 2

Theorem (Geck–Hiss 1991)

Assume that ℓ is good. Let $s \in \mathbf{G}^$ be a semisimple F^* -stable ℓ' -element. Then the series $\mathcal{E}(\mathbf{G}, s)$ forms a basic set for $\mathcal{E}_\ell(\mathbf{G}, s)$.*

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To each unipotent conjugacy class C of $\mathbf{G}$, one associates n_C projective modules Γ_i^C for $1 \leq i \leq n_C$, called the **generalised Gelfand–Graev representations** (GGGRs).

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*We call this class the **wave front set** of V .*

This theorem is only known when p is **good**.

Step 3: Kawanaka characters

 C_1 C_2 C_k

Fix an order on the k unipotent conjugacy classes $C_1, \dots, C_k$ of $\mathbf{G}$ such that $i < j$ if $C_i \subseteq \overline{C_j}$.

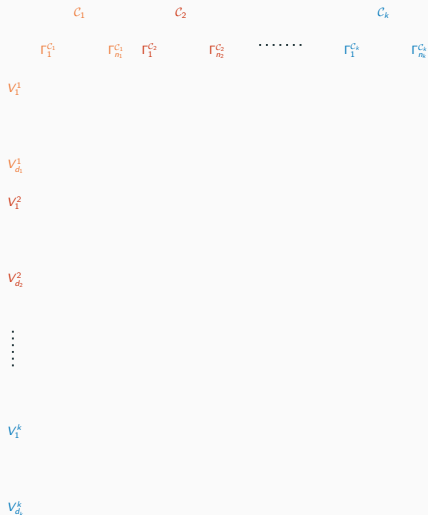
Step 3: Kawanaka characters

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Order the candidates $V_1, \dots, V_n$ according to their wave-front set.

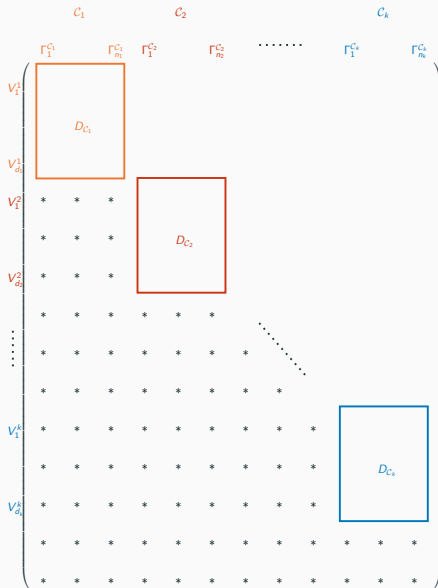
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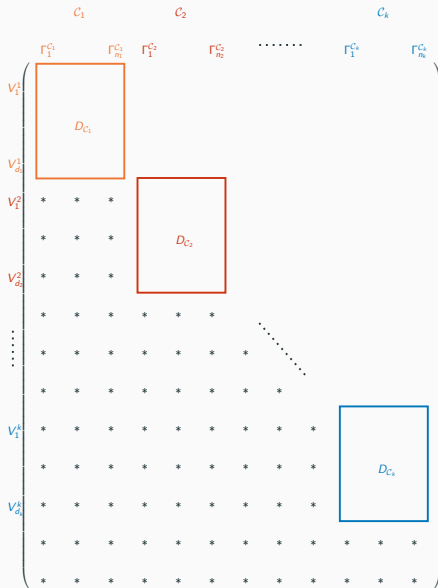
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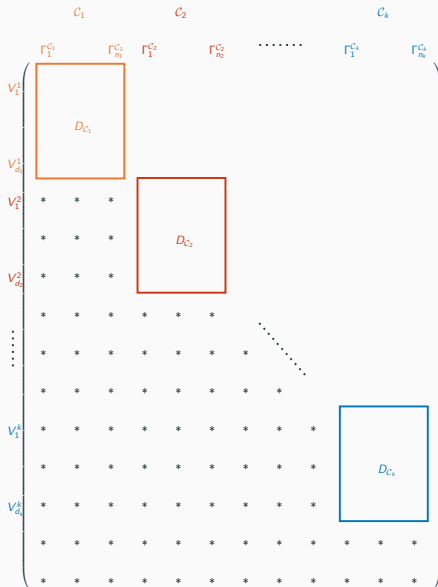


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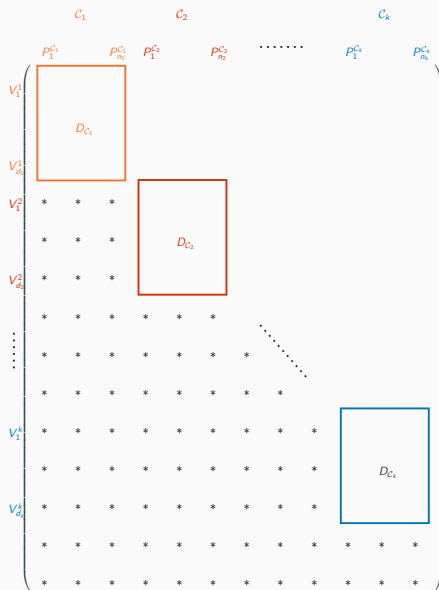
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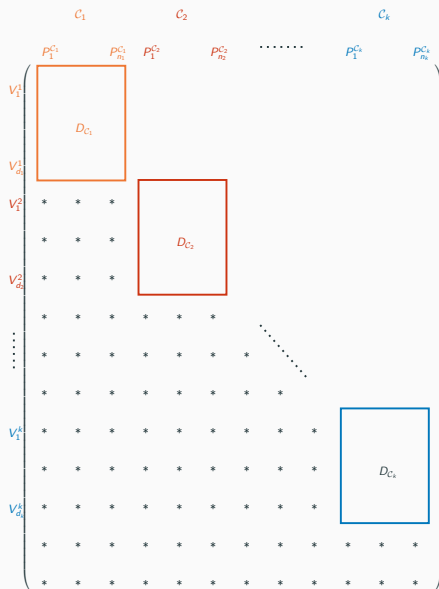
Solution: decompose the GGGRs into **Kawanaka** modules.

Step 4: unitriangularity



In most cases, we have enough candidates for the projective modules $P_1, \dots, P_n$.

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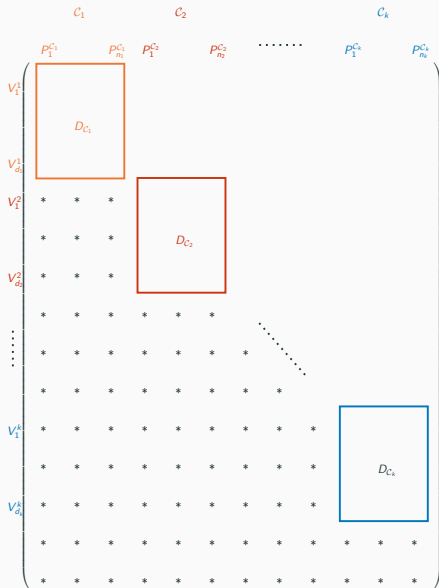


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For each unipotent conjugacy class C , check that the matrix D_C is unitriangular.

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We need the characters values of the ordinary modules in the basic set.

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Character sheaves are $\mathbf{G}$ -equivariant irreducible perverse sheaves on $\mathbf{G}$.

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Unipotent support	Unipotent support/wave front set

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Let $\mathcal{A} \in \hat{\mathbf{G}}$. We want the value of $\mathcal{A}_{(su)_G}$ for $s \in \mathbf{G}$ semisimple, $u \in C_{\mathbf{G}}^{\circ}(s)$ unipotent, and $(u)_G$ is the unipotent support of $\mathcal{A}$.

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The values of character sheaves at unipotent conjugacy classes are computable.

This is due to the **generalised Springer correspondence**.

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If $\mathbf{G}$ is simple of adjoint type and p is good for $\mathbf{G}$, then the value of $\mathcal{A}_{(su)_{\mathbf{G}}}$ is “almost” computable.

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$\alpha(\mathcal{A}, X)$ is approximately $\langle Y, \text{Res}_{W_{\mathfrak{m}}}^{W_{\mathbf{L}_0}^{\mathbf{G}}} \text{Ind}_{W_{\mathbf{L}_0}^{\mathbf{H}}}^{W_{\mathbf{L}_0}^{\mathbf{G}}}(X) \rangle$ with a twist.

Main results

Recall that $\ell \neq p$.

Main theorem (R. 2025)

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Additional Result (R.)

If $\mathbf{G}$ is of type G_2 , F_4 , or E_6 , the isolated ℓ -blocks are also lower-unitriangular.

Thank you!

Comments on the results/proofs

- The condition p good is crucial for the GGGRs.
- Some cases are a consequence of a result of Geck–Hézar (2008).
- In a lot of cases, there is no need to compute the values of all character sheaves at all mixed conjugacy classes, but only at the unipotent support.
- For the isolated blocks, Kawanaka characters are sometimes not enough.

Solution: apply Harish-Chandra theory and the knowledge that the decomposition matrix of the Levi subgroup is lower-unitriangular.

Next questions to consider

- Unitriangularity of decomposition matrices for the isolated blocks of type E_6 , E_7 and E_8 .
- Unitriangularity of decomposition matrices in the disconnected centre case.
- Properties of basic sets, e.g. stable under group automorphisms.
- Characteristic functions of character sheaves at mixed conjugacy classes.

References

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